

The Holistic Storage of Verb+*Up* Phrases in Text-based and Audio-based Language Models

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Abstract

One of the most central aspects of language processing is the ability to trade off between stored representations and abstract knowledge: one must retrieve stored representations, but also generate novel ones by applying productive rules. While recent work has examined abstract knowledge in language models, holistic storage has received far less attention. We probe internal representations in both text-based LLMs and an ASR model, testing whether V+*up* phrasal verbs develop distinct representations as a function of frequency and predictability. All models show evidence of holistic storage driven by frequency and predictability, further supporting usage-based theories of language.

1 Introduction

A central debate in linguistics concerns how humans trade off between computation and storage (Houghton, 2025a,b; Houghton and Morgan, 2023, 2024; Kapatsinski et al., 2009; Morgan and Levy, 2015, 2016, 2024; Stemberger and MacWhinney, 2004, 1986). Computation refers to applying abstract knowledge to generate new representations; for example, deriving *wugs* from *wug* via a productive pluralization rule. Storage refers to retrieving a whole representation from memory rather than computing it, such as accessing a holistic representation of a common phrase (e.g., *I don't know*) despite being able to, in principle, generate the phrase compositionally. Both mechanisms are clearly at work in language learning and processing, but the factors that drive what forms are produced via computation and what forms are stored and retrieved holistically remains poorly understood.

1.1 Computation vs Storage in Humans

There is an abundance of evidence for abstract knowledge in language use by humans. Children generalize morphological rules productively to novel words they have never encountered (Berko,

1958), and priming studies show that sentences sharing syntactic structure or semantically related words facilitate each other's processing, implying that something abstract is shared across their representations (Bock, 1986; Meyer and Schvaneveldt, 1971). Similarly, when ordering novel or low-frequency binomials, humans rely on abstract preferences (e.g., preferring the shorter word first) rather than simply producing the more frequent ordering (Morgan and Levy, 2016).

Storage has had a more controversial history. Early accounts held that only irregular forms (e.g., *went*) were stored holistically and that any compositionally derivable form was derived via computation (Pinker and Ullman, 2002). Evidence against this view has since accumulated (Bybee and Scheibman, 1999; Houghton, 2025b; Kapatsinski et al., 2009; Morgan and Levy, 2016; Stemberger and MacWhinney, 2004). Stemberger and MacWhinney (2004) showed that inflection errors are less common for high-frequency words, suggesting holistic storage even for regularly derived forms. Bybee and Scheibman (1999) demonstrated that *don't* is phonetically more reduced in high-frequency phrases like *I don't know* than in lower-frequency phrases like *I don't go*; if *don't* always had the same representation, such context-specific reduction would be difficult to explain.

Processing studies have provided converging evidence. Morgan and Levy (2016) found that while human ordering preferences for low-frequency binomials are driven by abstract preferences, preferences for high-frequency binomials are driven by item-specific preferences, suggesting a frequency-dependent shift from computation to storage. Most directly relevant to the present study, Kapatsinski et al. (2009) used a button-pressing task, in which participants responded upon hearing *up* within a word or a V+*up* phrase, and found that reaction times were U-shaped across frequency: faster for medium-frequency V+*up* phrases than

low-frequency ones, but slower again for high-frequency phrases. Further, Houghton (2025b) found that this pattern holds for high-predictability phrases (V+*up* phrases where *up* is likely to appear given the verb), suggesting that high-frequency and high-predictability V+*up* phrases are represented holistically.

1.2 Computation vs Storage in LMs

Whether language models exhibit analogous computation-storage tradeoffs to humans has become an active area of inquiry. On the computation side, results are mixed: some studies find that models learn abstract generalizations not present in training (Misra and Mahowald, 2024; Yao et al., 2025), while others find that models fail to use abstract knowledge where humans do, such as morphological generalization to novel words (Haley, 2020). On the storage side, there is no doubt that models rely heavily on memorization (McCoy et al., 2023). Indeed, item-specific frequency effects have been documented even in tasks where humans show abstract preferences (Houghton et al., 2025).

Despite these findings, it remains unclear as to whether LLMs develop holistic phrasal representations in a manner similar to humans. If they do, this would suggest that holistic storage is a natural consequence of learning from distributional patterns in the language, requiring no explicit storage mechanism, and lending support to usage-based accounts of how computation and storage interact. Audio-based models are especially well-suited to address this question because a good deal of the evidence for holistic storage comes from listening paradigms. It is thus important to understand whether ASR models, not just LLMs, show evidence of holistic storage; yet holistic storage has never been examined in any ASR model (though see Pluth et al., 2026, for related mechanistic interpretability work on Whisper), and has rarely been examined in any modality or in models trained on human-comparable amounts of data.

1.3 Present Study

The present study addresses this gap. We probe internal representations in text-based language models that were trained on an amount of data comparable to humans (BabyLMs),¹ a large language

model (OLMo-3 7B), and an audio-based speech recognition model (Whisper-small). These models were chosen to help illuminate effects of training size, number of parameters, and modality (speech vs text). In order to probe their representations, we trained logistic classifiers to detect the embedding of *up* as a preposition (outside of V+*up* contexts), then tested the classifier on V+*up* phrases of varying frequency and predictability. If these models develop holistic representations for V+*up* phrases, the representation of *up* in high-frequency and high-predictability V+*up* phrases should diverge from that of the preposition *up* more than in lower-frequency and lower-predictability phrases, resulting in lower logit scores by the classifier. Our specific contributions are:

- We train and release three open-access autoregressive models trained on the BabyLM v3 corpus (Charpentier et al., 2025), checkpointed every 20M tokens, to facilitate future research on human-scale language learning.²
- We show that holistic phrasal storage emerges in both text-based LLMs and an ASR model, establishing that frequency- and predictability-driven holistic representations arise even from models trained on an amount of data comparable to humans, and even across modalities.
- We show that frequency effects on phrasal storage are robust across model sizes, but predictability effects strengthen with scale, suggesting that sensitivity to co-occurrence statistics beyond raw frequency requires greater representational capacity.

2 Model Training

In order to examine holistic storage in models that have seen human-comparable amount of data, we trained three language models on the BabyLM v3 corpus (Charpentier et al., 2025), a 150M-token dataset designed to be more reflective of the scale and quality of the language that humans receive.³

a little less than half the tokens that an average college-aged human has experienced.

²A HuggingFace link is omitted here to preserve double-blind review; model weights and checkpoints will be released publicly upon publication.

³Though it is worth noting, as has been pointed out before (e.g., Houghton, 2025b), that it may be misleading to compare the tokens that LLMs receive to the “tokens” that humans receive, since humans encounter language in a context-rich environment while LLMs see only the raw text.

¹The model was trained on 150 million tokens. The average college-aged human experiences approximately 350 million words (Levy et al., 2012) so the model is trained on

All three models follow the OPT decoder-only transformer architecture (Zhang et al., 2022), with 125M, 350M, and 1.3B parameters, respectively.

Prior to training, we fit a byte-pair encoding (BPE) tokenizer directly on the BabyLM training corpus, yielding a vocabulary of 8,192 subword types. This tokenizer was shared across all three model sizes, ensuring that cross-model comparisons are not confounded by differences in tokenization. A full description of the model training is included in Appendix A; training loss curves, final gradient norms, and validation perplexity confirming that all three models converged are reported in Appendix B.

3 Experiment 1: UP Independently

Experiment 1 tests whether LLMs develop holistic phrasal representations analogous to those proposed by usage-based theories (e.g., Houghton, 2025b), using classifiers trained on prepositional *up* representations and applied to V+*up* phrases varying in frequency and predictability. We investigate this across OLMo-3 7B and the three BabyLMs (OPT-125M, 350M, and 1.3B).

3.1 Methods

For each model, we extracted the representation of the token *up* from each hidden layer of the models for each sentence. Using these representations, we trained a separate logistic regression classifier for each layer to distinguish prepositional *up* (which did not occur in V+*up* phrases in the training of the classifier) from other tokens in the same sentence (these other tokens did not contain the segment *up* in any capacity). The trained classifier was then tested on a held-out test set comprising V+*up* phrases, and for each sentence the classifier returned a logit score reflecting the predicted probability that the representation of *up* in the V+*up* phrase resembled the standalone, prepositional class.

Frequency for each V+*up* type is operationalized as its raw corpus count, log-transformed. Letting $c_{vup} = \text{count}(V+up)$:

$$\text{log-frequency} = \log(c_{vup}) \quad (1)$$

Predictability is operationalized as the log-odds ratio of V+*up* occurrences to V not followed by *up* in the corpus.⁴ Letting $c_V = \text{count}(V)$:

⁴Note that this is mathematically equivalent to taking the logit of the conditional probability of *up* given the verb.

$$\text{log-predictability} = \log\left(\frac{c_{vup}}{c_V - c_{vup}}\right) \quad (2)$$

Counts were derived from the training corpus of each model: the BabyLM V3 dataset (Charpentier et al., 2025) for the BabyLM models, and Dolma v1.7 (queried via the infini-gram API; Liu et al. 2024) for OLMo-3 7B. Although OLMo-3 7B was trained on Dolma 3, Dolma v1.7 is the most recent snapshot indexed by infini-gram and draws from the same underlying sources, making it a reasonable approximation of OLMo-3 7B’s training distribution.⁵

3.1.1 Classifier Training

The classifier was trained to distinguish the language models’ representations of the preposition *up* from representations of other tokens in the same sentence. Positive training examples consisted of 1,000 occurrences of *up* which occurred in sentences strictly as a preposition,⁶ drawn from sentences in the C4 corpus (Raffel et al., 2020). Because *up* is the only word among these positive examples, this set is simply 1,000 different sentences containing that same word, not 1,000 different words. Negative examples were 1,000 tokens randomly selected from the same sentences, restricted to tokens whose decoded string consists entirely of alphabetic characters (no numbers, punctuation, or special characters), and excluding the preposition *up* itself and any token containing *up* as a substring. The validation set was drawn from the same pool of sentences (a non-overlapping subset), with token positions resolved per model’s tokenizer; exact counts vary slightly by tokenizer (Appendix E). The classifiers for all models were trained on the same underlying sentences.⁷

The test set comprised V+*up* phrases (e.g., *pick up*) with at least 20 occurrences in the corpus; up to 20 sentences were sampled per type. Because the BabyLM corpus is smaller than Dolma, fewer V+*up* types attain a valid (non-zero) predictability estimate; only types with a valid estimate are

⁵The relative frequencies of common English phrasal verbs are unlikely to differ substantially between Dolma versions, as both are large-scale web-text corpora of similar composition.

⁶We used a morphological parser to filter out sentences in which *up* was not tagged as a preposition. The filter is deliberately conservative: it requires the unambiguous dependency label prep.

⁷Across all layers, models, and experiments, every classifier achieved above 94% accuracy on its held-out validation set. A full per-layer breakdown is reported in Appendix C.

included in the analyses. Full item-level statistics are reported in Appendix D (Table 6), but broadly speaking there were 4,081 unique *V+up* types for OLMo-3 and 1,039 for the BabyLM models. Classifier training and validation split sizes are shown in Appendix E (Table 7).

3.1.2 Analyses

In order to examine the effects of frequency and predictability on the representations of *up* in *V+up* phrases, we implemented a Bayesian mixed-effects regression model using the *brms* package (Bürkner, 2017). For each statistical analysis, the outcome variable was the logit score returned by the classifier on each test item. We included fixed-effects for frequency and predictability, both of which were centered and scaled (denoted c_log_freq and c_log_predic below). We also included random intercepts for phrasal verb type. Weak, uninformative priors were included on each fixed-effect. The model syntax is included below:

$$\text{logit} \sim c_log_freq \times c_log_predic + (1 \mid \text{verb_up}) \quad (3)$$

Frequency and predictability are correlated to some degree, but because both are included in the same model, each coefficient reflects effects above and beyond the other, and variance inflation factors remain low across all models (Appendix D).

Separate statistical models were fit for each language model at the final hidden layer. For each coefficient we report the posterior mean, standard error, 95% credible interval (CI), and the proportion of posterior samples greater than zero ($\% > 0$). We consider an effect to be meaningful if the 95% CI does not contain zero.⁸

In order to examine the difference in representations across hidden layers, we additionally fit a generalized additive model (GAM, Wood, 2017). As with the Bayesian analyses, the dependent variable was the classifier logit. Specifically, two models were fit separately, one for frequency and one for predictability, each with a tensor product smooth over the predictor and hidden layer index (with a random intercept for verb):

⁸Though as Houghton et al. (2024) point out, unlike frequentist statistics, Bayesian analyses don't force us to commit to a binary of significance vs. non-significance, and the percentage of samples greater than zero can be interpreted in a continuous manner.

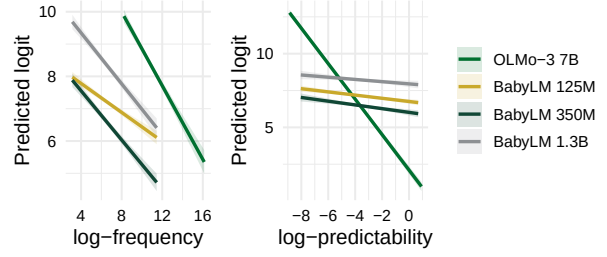


Figure 1: Final-layer brms predicted logit by frequency (left) and predictability (right) for all models (UP independently). Shading indicates 95% CIs.

$$\begin{aligned} \text{logit} &\sim \text{te}(\log_freq, \text{layer}) \\ &\quad + s(\text{verb_up}, \text{bs} = 're') \\ \text{logit} &\sim \text{te}(\log_predic, \text{layer}) \\ &\quad + s(\text{verb_up}, \text{bs} = 're') \end{aligned} \quad (4)$$

3.2 Results

Full numerical results are reported in Appendix E (Table 8 for the Bayesian model and Table 9 for the GAM); they are visualized in Figure 1 and Figure 2. Figure 8 (Appendix E) shows the layer-by-layer difference between high- and low-predictor items more directly.

For all four LLMs, there was a negative effect of both frequency and predictability on the classifier's logit score. In other words, both frequency and predictability result in the representation of *up* in *V+up* phrases being less similar to the prepositional representation of *up*. We also observe a negative frequency-by-predictability interaction for all three BabyLM models but not for OLMo-3 7B (Table 8).

Additionally, the by-layer analysis demonstrates that predictability's divergence begins to appear early on in the larger models (specifically, OLMo-3 7B and BabyLM 1.3B), while taking longer to emerge in the smaller models.

3.3 Discussion

Experiment 1 found that the logits from a classifier trained on the preposition *up* representations decreased for high-frequency and high-predictability *V+up* phrases. Additionally, the effect of predictability appears to be weaker in the smaller models. These results parallel those of Kapatsinski et al. (2009) and Houghton (2025b), who examined this in humans. In Experiment 2, we more closely parallel these studies by also examining a probe trained to identify the *up* segment, regardless of semantics.

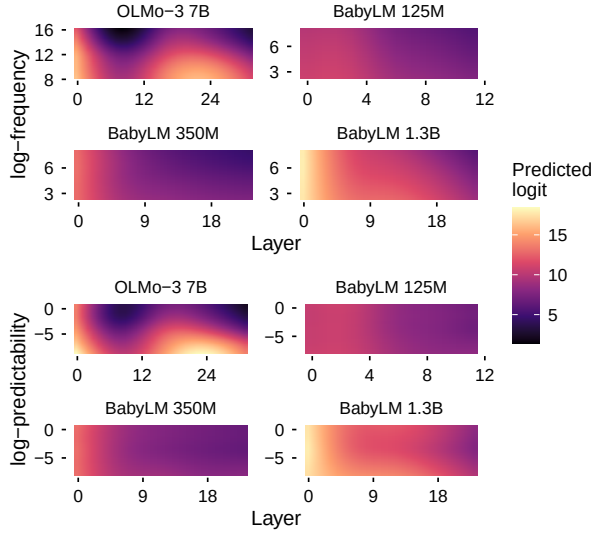


Figure 2: GAM-predicted logit by layer and predictor for all models (UP independently). Top: frequency; bottom: predictability.

4 Experiment 2: UP as Subword

One possible criticism of the results of Experiment 1 is that the lower logit score for frequent/predictable V+*up* phrases could reflect semantic bleaching or polysemy rather than holistic storage. To address this concern, we replicate the design of Experiment 1 with one key change: rather than training the classifier on the preposition *up* exclusively, we additionally train it on instances of *up* embedded within a larger word (e.g., *update*), requiring the model to identify the sequence *up* regardless of whether it functions as a preposition. This tests whether the frequency and predictability effects observed in Experiment 1 generalize to the *up*-segment more broadly. This design is also arguably more faithful to Kapatsinski et al. (2009) and Houghton (2025b), where participants were tasked with recognizing the segment *up* in general.

4.1 Methods

The procedure was identical to Experiment 1 (same models; classifiers trained and evaluated at each hidden layer; same test set, outcome variable, and statistical approach) with one difference: the classifier was trained on a broader set of positive examples that included instances of *up* embedded within a larger word (e.g., *update*, *upon*).

4.1.1 Classifier Training

The training set for this experiment combined two types of positive examples: 1,000 occurrences of the preposition *up* (identical to Experiment 1) and

1,000 occurrences of *up* embedded within a larger word (e.g., *update*, *upon*, etc). Critically, the subword positives were restricted to **unique word types**: each *up*-containing word contributed exactly one instance, so the classifier could not learn to recognize a high-frequency form like *upon* from repeated exposure. Negative examples (1,000 drawn from each sentence pool) were tokens consisting entirely of alphabetic characters that did not contain *up* as a substring. The validation set was constructed by the same procedure (targeting 1,000 positive, 1,000 negative per sentence pool; exact counts vary by tokenizer). As in Experiment 1, each model used the same underlying sentences with their own tokenizers. The test set was identical to Experiment 1. Classifier training/validation split sizes are shown in Appendix F (Table 10).

4.1.2 Analyses

Analyses were identical to those of Experiment 1, except using the logits from the classifiers trained in the current experiment.

4.2 Results

Full numerical results are reported in Appendix F (Table 11 for the Bayesian model and Table 12 for the GAMs); they are visualized in Figure 3 and Figure 4. Figure 9 (Appendix F) shows the layer-by-layer difference more directly. Frequency effects replicate Experiment 1 across all four models. Predictability effects are more variable: BabyLMs show absent or positive effects, OLMo-3 7B shows progressively stronger negative effects, consistent with predictability sensitivity increasing with scale (though it's unclear whether it's due to the scale of data or scale of the model). Similar to Experiment 1, there is also a negative interaction effect between frequency and predictability, however this time only credible for the two smaller BabyLMs (125M and 350M). The by-layer patterns similarly show a progressively stronger effect of frequency for later layers relative to earlier layers, while the effect of predictability follows a similar pattern for OLMo, but not for the BabyLMs. For the BabyLMs, the effect of predictability either stays relatively constant across layers or, in the case of the 1.3B model, becomes negative around layer 5 and then slowly grows positive again.

4.3 Discussion

The results of Experiment 2 mostly replicate those of Experiment 1. Frequency shows a strong, con-

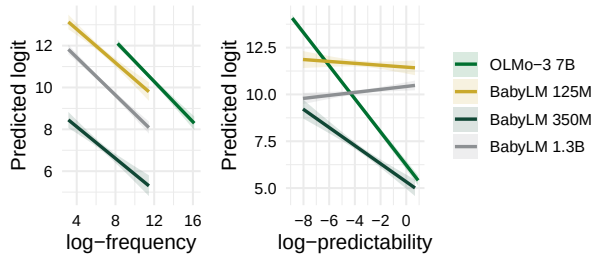


Figure 3: Final-layer brms predicted logit by frequency (left) and predictability (right) for all models (UP as subword). Shading indicates 95% CI.

sistent effect such that the representation of *up* in high-frequency V+*up* phrases diverges from representations of *up*, even for a probe trained to identify *up* as a segment, regardless of the semantics. The effect of predictability seems to emerge in larger models, suggesting that sensitivity to predictability may require more representational capacity than sensitivity to frequency. The results also mitigate concerns that Experiment 1’s findings reflect polysemy or semantic bleaching: Experiment 2’s positive class is defined purely orthographically, regardless of semantics, yet the effects of frequency and predictability largely replicate.

5 Experiment 3: Whisper (ASR Model)

In Experiment 3, we apply the same classifier approach to Whisper-small, an ASR model trained on spoken audio rather than written text. Whisper differs from the models in Experiments 1 and 2 in both its training modality and its encoder-decoder architecture. Examining Whisper allows us to ask whether the frequency and predictability effects on phrasal representations generalize beyond text-based models to representations learned from speech.

5.1 Methods

The procedure mirrored Experiments 1 and 2, with two key differences: (1) the model was Whisper-small; and (2) the stimuli were spoken audio segments rather than written sentences. For each segment, we ran the audio through Whisper and extracted the hidden-state representation of *up* at each layer of the encoder and decoder separately. A logistic regression classifier was trained independently for each component (encoder, decoder) to differentiate Whisper’s embeddings of *up* as a standalone preposition from non-*up* tokens (see Classifier Training below), then applied to spoken V+*up* phrasal verb segments; an extension of this Exper-

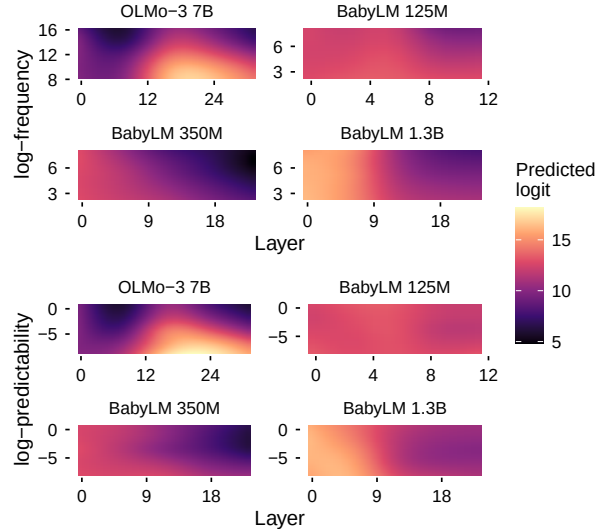


Figure 4: GAM-predicted logit as a function of transformer layer (x-axis) and predictor value (y-axis) for each model (UP as subword). Top row: log-frequency; bottom row: log-predictability. Color encodes the predicted logit score (random effects excluded).

iment with the probe trained on words containing *up* as well is presented separately as a robustness check (Appendix H).

5.1.1 Classifier Training

The Whisper experiment used spoken language data drawn from a subset of the GigaSpeech corpus (Chen et al., 2021) (corpus construction statistics in Appendix G). Word-level timestamps within each segment were obtained using WhisperX forced alignment. In order to classify *up* accurately, we used the ground-truth transcripts of the audio segments rather than the Whisper transcribed ones (to avoid any potential confound). We classified *up* by running spaCy part-of-speech tagging on the ground-truth transcript: if the token immediately preceding *up* was tagged as a VERB, the instance was labelled V+*up* (e.g., *pick up*, *clean up*). This classification procedure mirrors the one used to identify the preposition *up* in Experiment 1.

The **training and validation sets** used 1,000 preposition *up* instances each as positive examples, paired with 1,000 randomly selected non-*up* words from the same segment as negative examples.

The **test set** comprised V+*up* types with at least 5 occurrences in the audio dataset (lower than Experiments 1 and 2’s threshold of 20, reflecting the relative sparsity of the audio domain compared to text corpora), with up to 20 instances per type. Corpus frequency and predictability were drawn from Dolma v1.7 (via infinigram), identical to OLMo-3

7B. After filtering to items with a valid predictability estimate, 1,460 unique V+*up* types were retained. Split sizes are included in Appendix G (Table 13) and item-level statistics in Appendix D (Table 6).

5.1.2 Analyses

Analyses followed the same procedure as Experiments 1 and 2, using the same predictor operationalizations (Equation 1, Equation 2). Because audio segments, unlike text tokens, can vary in acoustic duration, we additionally included the duration of the *up* segment (*c_duration*, centered and scaled) as a covariate in the joint model, to control for the possibility that frequent or predictable V+*up* phrases are simply phonetically reduced:

$$\text{logit} \sim c_{\text{log_freq}} \times c_{\text{log_predic}} + c_{\text{duration}} + (1 \mid \text{verb_up}) \quad (5)$$

The brms models were fit separately for Whisper’s encoder and decoder at the final hidden layer. The by-layer GAMs likewise included duration as an additional covariate:

$$\begin{aligned} \text{logit} &\sim \text{te}(\text{log_freq}, \text{layer}) + c_{\text{duration}} \\ &\quad + s(\text{verb_up}, \text{bs} = 're') \\ \text{logit} &\sim \text{te}(\text{log_predic}, \text{layer}) + c_{\text{duration}} \\ &\quad + s(\text{verb_up}, \text{bs} = 're') \end{aligned} \quad (6)$$

5.2 Results

Full numerical results are reported in Appendix G (Table 14 for the Bayesian model and Table 15 for the GAM model); they are visualized in Figure 5 and Figure 6 below.⁹ Figure 10 (Appendix G) shows the layer-by-layer difference between high- and low-predictor items more directly.

The results for Whisper diverge between the two components. The decoder shows a negative effect of both frequency and predictability, controlling for the duration of the *up* segment, consistent with the pattern found throughout the paper; both effects remain credible even with duration included in the model, arguing against phonetic reduction as an

⁹As a further robustness check, we replicated this design with a broader positive training class that also included *up* embedded within other words, following Experiment 2’s sub-word criteria adapted for audio. The decoder’s results parallel those reported here; the encoder’s predictability effect likewise replicates, and its frequency effect replicates in direction but emerges as credible (Appendix H).

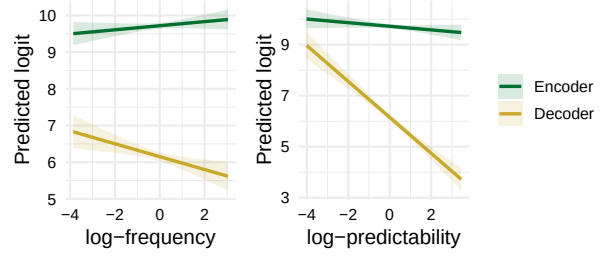


Figure 5: Final-layer brms predicted logit by frequency (left) and predictability (right) for Whisper encoder and decoder. Shading indicates 95% CI.

alternative explanation. The encoder shows a different pattern: predictability has a negative effect whose 95% credible interval marginally includes zero (94.5% of the posterior mass falls below zero), while frequency shows a marginally positive effect (91.0% of the posterior mass above zero). Duration itself has a credible negative effect in the encoder and a credible positive effect in the decoder.

The by-layer analysis shows the encoder’s effects are variable across layers: predictability stays negative throughout, most pronounced in the middle layers, while frequency stays negative through the early and middle layers before rising sharply in the final third of the network and crossing to a small positive difference by the final layers, consistent with the final-layer joint model’s near-zero, non-credible frequency effect reported above. In the decoder, frequency drops sharply from near zero to strongly negative by the middle layers and remains strongly negative through the final layer, while predictability decreases steadily and monotonically across the entire network, reaching its most negative value at the final layer (Table 15).

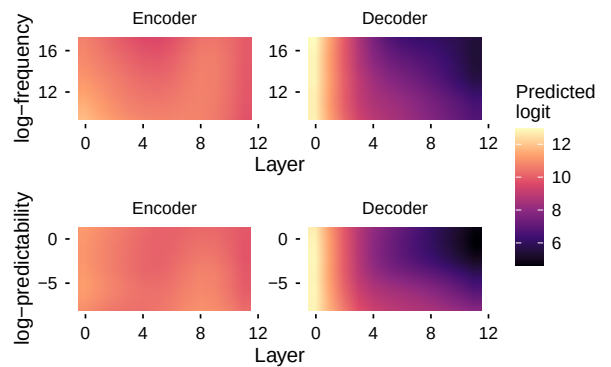


Figure 6: GAM-predicted logit by layer and predictor for Whisper encoder and decoder. Top: frequency; bottom: predictability. Color = logit; darker = lower logit (less similar to the preposition *up*).

5.3 Discussion

The results of Whisper demonstrate an interesting similarity between audio speech recognizers and text-based language models: both the encoder and the decoder of Whisper show similar patterns to the LLMs for predictability, and the decoder shows the same pattern for frequency. In the encoder, however, frequency’s relationship is positive rather than negative. The by-layer trajectory offers a possible explanation, distinct from phonetic reduction (which the duration covariate already controls for at every layer): the encoder’s frequency effect is negative in early layers and crosses over to positive by the final layers (Figure 10). One interpretation is that the encoder’s early layers retain acoustically fine-grained detail, on which frequent instances of *up* may simply be more acoustically variable than the relatively consistent prepositional form, while later layers progressively abstract over this acoustic variability. Under this view, the crossover reflects the encoder collapsing surface acoustic differences. The same crossover pattern holds under subword training, where it is even more pronounced (Appendix H), consistent with growing acoustic invariance across layers.

6 Conclusion

The present study demonstrates that both text-based LLMs and ASR models represent frequent/predictable *V+up* phrases holistically, with predictability effects emerging as a function of model size and robust across differences in tokenization and modality, though the ASR model’s encoder shows a more complex relationship with frequency. Notably, this holds even in models trained on an amount of data comparable to humans, suggesting that holistic storage does not require the massive training corpora typical of modern LLMs. Further, the effect of predictability is clearer and emerges in earlier layers for larger models compared to smaller ones, suggesting that predictability effects emerge from greater representational capacity.

The present results parallel those from the human behavioral data: just as listeners are slower to recognize *up* within frequent or predictable *V+up* phrases (Houghton, 2025b; Kapatsinski et al., 2009), a lower classifier logit reflects *up*’s representation in *V+up* phrases becoming less similar to that of the standalone preposition (as a function of frequency/predictability). Under a purely com-

positional account, the representation of *up* should be static regardless of context, whether in *sketch up* or *pick up*. Under usage-based accounts, by contrast, the *V+up* phrase is compositional at low frequency/predictability, like *sketch up*: *up* contributes its typical meaning, so its representation should resemble the standalone preposition’s. As frequency/predictability rise, the phrase becomes progressively less compositional and *up*’s representation diverges accordingly; holistic storage is the term for this divergence at its most extreme, as in *pick up*. The data supports this graded account specifically: the classifier’s confidence that *up* resembles its prepositional form is highest for low-frequency/predictability items and decreases as frequency/predictability rise, rather than uniform across all *V+up* items, as we would expect if the mere presence of a *V+up* context, rather than frequency or predictability specifically, drove the change. This pattern also cannot be attributed to the tokenization method: in LLMs, the token for *up* is identical across all items, whereas each *up* token in Whisper’s audio input is acoustically distinct, yet the same divergence emerges in both. This therefore presents a mechanistic explanation for the slowdown seen in the human behavioral data.

Lastly, the present results suggest that one need not posit separate storage and abstraction mechanisms to account for the human behavioral data: if holistic storage and grammatical abstraction arise from the same process, the traditional distinction between stored items and productive rules may be a post-hoc description of a continuous representational landscape rather than a reflection of genuinely distinct cognitive processes. The gradient by-layer results support this view: rather than a sharp transition from compositional to holistic representations, we observe a progressive divergence that deepens across layers and strengthens with frequency and predictability, more consistent with storage-like behavior being one end of a continuum than with a strict division between stored and computed items.

Overall, the present study demonstrates another dimension along which the behavior of transformer models, across modalities, resembles that of humans, providing further evidence for usage-based theories: storage-like representations emerge gradually as a function of frequency and predictability, arising even in models with no explicit storage mechanism and, for the BabyLMs, no more data than a human receives. More broadly, the re-

sults suggest that transformer architectures offer a productive framework for investigating not just whether human-like storage patterns emerge, but *how* and *when* they do so, at a level of mechanistic detail that behavioral data alone cannot provide.

7 Limitations

The primary limitation is that we examined only one construction (V+*up* phrases) in one language (English). We chose this construction deliberately: V+*up* phrasal verbs are among the most frequent and productive multi-word units in English, and they are the same construction examined in the human psycholinguistic literature this study builds on (Houghton, 2025b; Kapatsinski et al., 2009), which keeps the model-human comparison as direct as possible. The tradeoff is depth rather than breadth: rather than testing many constructions in a single model, we examined this one construction as thoroughly as our resources allowed, across three model families, two training modalities (text and speech), models trained on both human-scale and web-scale data, models varying in number of parameters, and every hidden layer rather than only the final one. Whether the same pattern holds for other multi-word constructions or other languages remains an open question, and one we see as a natural next step rather than a gap that undermines the present findings. We also only examined representations at the final checkpoint; however, we look forward to examining the emergence of storage as a function of training dynamics in the future.

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A BabyLM Model Training

Models were trained from random initialization for 20 epochs using the AdamW optimizer with fused weight updates and bfloat16 mixed precision. The learning rate was set to 3×10^{-4} for the 125M model and 1×10^{-4} for the 350M and 1.3B models, each preceded by a linear warmup over 10% of total training steps. Training was distributed across two NVIDIA A100 80GB GPUs. Hyperparameter details are given in Table 1.

Parameter	OPT-125M	OPT-350M	OPT-1.3B
Hidden size	768	1,024	2,048
Attention heads	12	16	32
Layers	12	24	24
FFN intermediate size	3,072	4,096	8,192
Vocabulary size	8,192	8,192	8,192
Context length (tokens)	1,024	1,024	1,024
Batch size (per device)	400	50	100
Gradient accumulation	1	2	1
Learning rate	3×10^{-4}	1.5×10^{-4}	1×10^{-4}
Warmup steps	366	732	1,465
Training epochs	20	20	20
Optimizer	AdamW (fused)	AdamW (fused)	AdamW (fused)
Precision	bfloat16	bfloat16	bfloat16

Table 1: Hyperparameters for the three BabyLM models. All models share a BPE tokenizer (vocabulary size 8,192) trained on the BabyLM corpus. Training used NVIDIA A100 80GB GPUs with distributed data parallelism: two for the 125M and 1.3B, four for the 350M.

B BabyLM Model Convergence

All three models trained to completion over 20 epochs, with training loss decreasing monotonically throughout (Figure 7). Final gradient norms were small for all models (0.16, 0.44, and 0.38 for the 125M, 350M, and 1.3B models, respectively), and the learning rate decayed to zero at the end of training, consistent with full convergence under the scheduled warmup-then-decay regime. No signs of loss divergence or instability were observed.

The 1.3B model achieves the lowest training loss (1.86), as expected given its greater capacity, while the 350M model achieves the lowest validation perplexity (14.35) (Table 2).

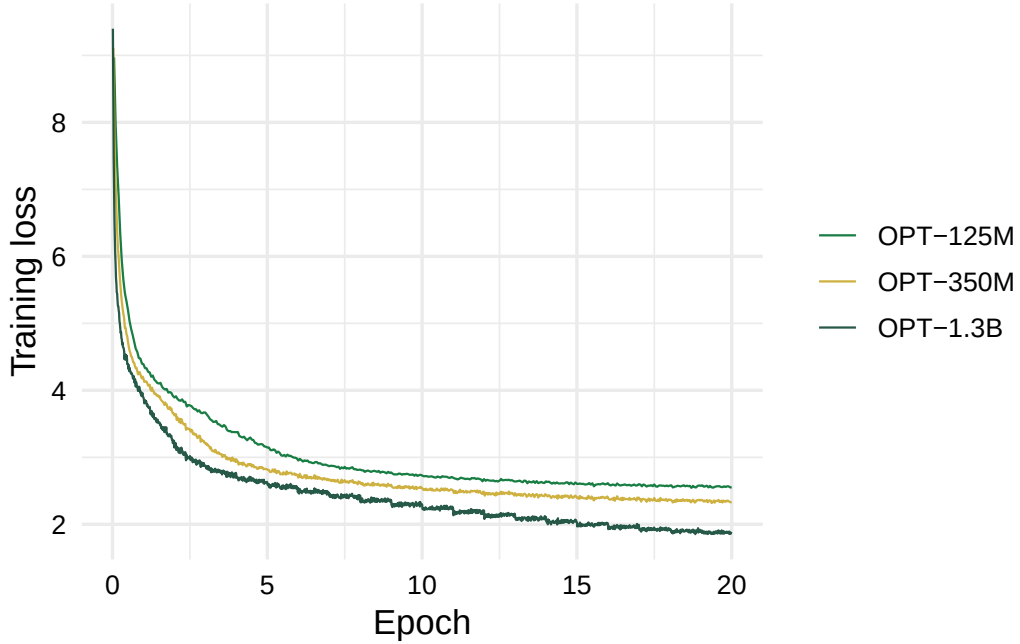


Figure 7: Training loss over 20 epochs for each BabyLM model. Loss is logged every 10 gradient steps.

Model	Val. loss	Perplexity
OPT-125M	2.82	16.81
OPT-350M	2.66	14.35
OPT-1.3B	2.72	15.18

Table 2: Final validation loss and perplexity for each BabyLM model, evaluated on the BabyLM development set after 20 epochs of training.

C Classifier Validation Accuracy by Layer

Tables 3–5 report held-out validation accuracy for every classifier trained in this study, broken down by layer, model, and condition (Experiment 1: independent *up*; Experiment 2: *up* as a subword within a larger word; Experiment 3: Whisper encoder/decoder). These are the per-layer values underlying the aggregate claim in the main text that every classifier, across all layers, models, and experiments, exceeded 95% validation accuracy, with one exception discussed there (the Whisper subword-replication decoder’s final layer). That exception comes from a separate robustness-check condition not included in Table 5 below (which reports Experiment 3 only); its own overall and subword-only per-layer accuracy is reported in Table 18 (Appendix H).

Table 3: OLMo-3 7B validation accuracy (%) by layer, Experiment 1 (independent *up*) vs. Experiment 2 (*up* as subword).

Layer	Exp. 1 (Indep.)	Exp. 2 (Subword)
0	100.0	99.7
1	100.0	99.7
2	100.0	99.7
3	100.0	99.7
4	99.9	99.5
5	100.0	99.5
6	99.9	99.5
7	100.0	99.3
8	99.9	99.2
9	100.0	99.1
10	100.0	98.9
11	100.0	98.9
12	100.0	98.6
13	100.0	98.9
14	100.0	98.5
15	100.0	98.6
16	100.0	98.6
17	99.9	98.6
18	100.0	98.5
19	100.0	98.4
20	99.9	98.4
21	99.9	98.3
22	99.8	98.3
23	99.8	98.3
24	99.7	98.0
25	99.6	98.1
26	99.6	97.8
27	99.6	97.9
28	99.6	97.5
29	99.6	97.6
30	99.3	97.2
31	99.4	97.4

Table 4: BabyLM validation accuracy (%) by layer and model, Experiment 1 (independent *up*) vs. Experiment 2 (*up* as subword). OPT-125M has 12 layers; “–” marks layers beyond a model’s depth.

Layer	OPT-125M		OPT-350M		OPT-1.3B	
	Exp. 1	Exp. 2	Exp. 1	Exp. 2	Exp. 1	Exp. 2
0	100.0	99.3	100.0	99.5	100.0	99.2
1	100.0	99.2	99.9	99.7	100.0	99.3
2	100.0	99.2	99.9	99.5	100.0	99.2
3	100.0	99.2	100.0	99.3	100.0	99.1
4	100.0	99.1	100.0	99.4	100.0	99.0
5	100.0	98.9	100.0	99.5	100.0	99.2
6	100.0	99.1	100.0	99.5	100.0	99.1
7	100.0	99.1	100.0	99.4	100.0	99.1
8	100.0	99.0	100.0	99.3	100.0	99.1
9	100.0	99.0	100.0	99.5	100.0	99.1
10	100.0	98.8	100.0	99.5	100.0	99.2
11	100.0	98.6	100.0	99.5	100.0	99.2
12	–	–	100.0	99.6	100.0	99.1
13	–	–	100.0	99.7	100.0	99.3
14	–	–	100.0	99.7	100.0	99.2
15	–	–	100.0	99.7	100.0	99.0
16	–	–	100.0	99.6	100.0	99.0
17	–	–	100.0	99.6	100.0	98.9
18	–	–	100.0	99.5	100.0	99.2
19	–	–	99.9	99.2	100.0	99.1
20	–	–	99.9	99.3	99.9	99.1
21	–	–	99.9	99.1	99.9	99.0
22	–	–	99.9	99.0	100.0	99.1
23	–	–	99.9	99.1	100.0	99.1

Table 5: Whisper validation accuracy (%) by layer and component, Experiment 3 (independent *up*). The Whisper subword replication (a robustness check, not a numbered experiment) is reported separately in Table 18 (Appendix H).

Layer	Encoder	Decoder
0	96.8	99.8
1	97.5	100.0
2	98.2	100.0
3	98.1	99.9
4	98.5	100.0
5	98.7	100.0
6	99.0	100.0
7	99.1	100.0
8	99.3	99.9
9	99.3	100.0
10	99.5	99.7
11	99.5	99.4

D Test Set Items

Frequency and predictability are correlated with one another to some degree ($r = 0.34$ for OLMo-3 7B, $r = 0.25$ for BabyLM, $r = 0.31$ for Whisper; see Table 6 for exact per-model values). Because both predictors are included in the same model rather than fit separately, each reported coefficient reflects the effect of that predictor above and beyond the other; the corresponding variance inflation factors are low across all models, indicating that this correlation does not meaningfully compromise the reliability of either estimate.

Model	N items	Med. freq	Freq range	Med. predic.	Predic. range	r	VIF
OLMo-3 7B	4081	43608	2–103,842,740	-5.38	-10.98–2.74	0.34	1.13
BabyLM	1039	34	10–9,963	-3.40	-10.38–1.99	0.25	1.07
Whisper encoder	1460	235806	960–103,842,740	-4.08	-10.98–2.74	0.31	1.10
Whisper decoder	1460	235806	960–103,842,740	-4.08	-10.98–2.74	0.31	1.10

Table 6: Test set statistics for all experiments. r and VIF give the Pearson correlation and variance inflation factor between log-frequency and log-predictability for that model’s test items.

E Experiment 1: UP Independently

Validation counts fall slightly below 1,000 for some models because the tokenizer does not always produce an isolated *up* token; for example, when *up* appears before punctuation and the tokenizer fuses the two into a single unit (e.g., *up,*), there is no isolable *up* position to extract, so that instance is excluded.

Model	Train pos	Train neg	Val pos	Val neg
OLMo-3 7B	1,000	1,000	1,000	1,000
BabyLM 125M	1,000	1,000	999	999
BabyLM 350M	1,000	1,000	999	999
BabyLM 1.3B	1,000	1,000	999	999

Table 7: Classifier training and validation split sizes for Experiment 1 (UP independently). Positive (pos) = preposition *up*; negative (neg) = other alphabetic token from the same sentence.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	8.25	0.05	[8.15, 8.36]	100.0
OLMo-3 7B	Freq	-1.00	0.06	[-1.11, -0.88]	0.0
OLMo-3 7B	Predic.	-2.72	0.05	[-2.83, -2.62]	0.0
OLMo-3 7B	Freq:Predic.	-0.01	0.05	[-0.12, 0.10]	41.9
BabyLM 125M	Intercept	7.13	0.04	[7.06, 7.20]	100.0
BabyLM 125M	Freq	-0.42	0.04	[-0.50, -0.35]	0.0
BabyLM 125M	Predic.	-0.20	0.04	[-0.27, -0.12]	0.0
BabyLM 125M	Freq:Predic.	-0.13	0.04	[-0.20, -0.05]	0.0
BabyLM 350M	Intercept	6.46	0.05	[6.36, 6.55]	100.0
BabyLM 350M	Freq	-0.71	0.05	[-0.81, -0.62]	0.0
BabyLM 350M	Predic.	-0.23	0.05	[-0.32, -0.13]	0.0
BabyLM 350M	Freq:Predic.	-0.12	0.05	[-0.22, -0.03]	0.6
BabyLM 1.3B	Intercept	8.21	0.05	[8.11, 8.31]	100.0
BabyLM 1.3B	Freq	-0.74	0.05	[-0.84, -0.63]	0.0
BabyLM 1.3B	Predic.	-0.14	0.05	[-0.24, -0.03]	0.5
BabyLM 1.3B	Freq:Predic.	-0.12	0.05	[-0.22, -0.01]	1.4

Table 8: Joint frequency $\times$ predictability model at the final layer for UP independently. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Predictor	EDF	F	p
OLMo-3 7B	Frequency	23.33	39583.62	< 0.001
BabyLM 125M	Frequency	15.32	16281.02	< 0.001
BabyLM 350M	Frequency	21.02	39055.63	< 0.001
BabyLM 1.3B	Frequency	16.76	29336.45	< 0.001
OLMo-3 7B	Predictability	23.40	45016.89	< 0.001
BabyLM 125M	Predictability	22.00	12479.55	< 0.001
BabyLM 350M	Predictability	22.56	35627.86	< 0.001
BabyLM 1.3B	Predictability	22.39	23854.74	< 0.001

Table 9: GAM tensor-product smooth summary for UP independently. EDF = estimated degrees of freedom; p = p-value for the te(predictor, layer) smooth term.

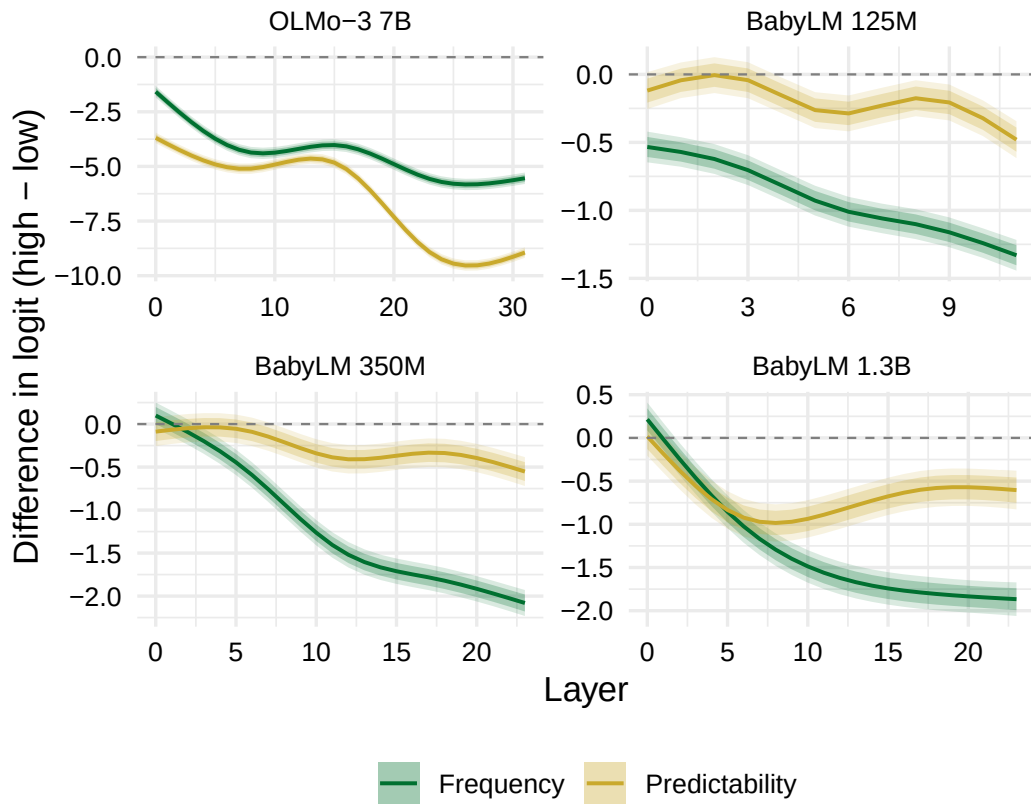


Figure 8: Layer-by-layer difference in GAM-predicted logit between high (90th percentile) and low (10th percentile) predictor values, for UP independently. Green = frequency effect; yellow = predictability effect. Inner and outer shaded ribbons show 80% and 95% confidence intervals derived from the GAM standard error. A negative difference indicates that high-predictor phrases yield a lower classifier logit (i.e., their representation of *up* is less similar to the prepositional class).

F Experiment 2: UP as Subword

Model	Train pos	Train neg	Val pos	Val neg
OLMo-3 7B	2,000 (1,000)	2,000	1,876 (876)	1,876
BabyLM 125M	2,000 (1,000)	2,000	1,582 (583)	1,582
BabyLM 350M	2,000 (1,000)	2,000	1,582 (583)	1,582
BabyLM 1.3B	2,000 (1,000)	2,000	1,582 (583)	1,582

Table 10: Classifier training and validation split sizes for Experiment 2 (UP as subword). Positive (pos) examples combine 1,000 prepositional *up* instances with up to 1,000 unique up-within-word types; parenthetical counts give the subword-specific portion (below 1,000 in validation where not every sampled instance yielded a resolvable token position). Negative (neg) = other alphabetic token from the same sentence pool. Val pos and Val neg are the balanced (truncated to equal class size) counts actually used for evaluation, as in Tables 7 and 13.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	10.75	0.05	[10.64, 10.84]	100.0
OLMo-3 7B	Freq	-0.84	0.05	[-0.95, -0.74]	0.0
OLMo-3 7B	Predic.	-2.00	0.05	[-2.10, -1.91]	0.0
OLMo-3 7B	Freq:Predic.	-0.05	0.05	[-0.15, 0.05]	14.6
BabyLM 125M	Intercept	11.63	0.08	[11.47, 11.79]	100.0
BabyLM 125M	Freq	-0.76	0.08	[-0.92, -0.59]	0.0
BabyLM 125M	Predic.	-0.09	0.08	[-0.25, 0.07]	13.0
BabyLM 125M	Freq:Predic.	-0.30	0.08	[-0.46, -0.13]	0.0
BabyLM 350M	Intercept	7.01	0.09	[6.84, 7.19]	100.0
BabyLM 350M	Freq	-0.71	0.09	[-0.89, -0.53]	0.0
BabyLM 350M	Predic.	-0.87	0.09	[-1.05, -0.68]	0.0
BabyLM 350M	Freq:Predic.	-0.19	0.09	[-0.37, -0.01]	1.7
BabyLM 1.3B	Intercept	10.15	0.05	[10.04, 10.25]	100.0
BabyLM 1.3B	Freq	-0.84	0.05	[-0.95, -0.74]	0.0
BabyLM 1.3B	Predic.	0.14	0.05	[0.04, 0.25]	99.6
BabyLM 1.3B	Freq:Predic.	-0.03	0.05	[-0.13, 0.08]	30.4

Table 11: Joint frequency $\times$ predictability model at the final layer for UP as subword. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Predictor	EDF	F	p
OLMo-3 7B	Frequency	23.36	44489.41	< 0.001
BabyLM 125M	Frequency	16.98	485.69	< 0.001
BabyLM 350M	Frequency	21.10	9188.77	< 0.001
BabyLM 1.3B	Frequency	19.74	16171.80	< 0.001
OLMo-3 7B	Predictability	23.12	48483.14	< 0.001
BabyLM 125M	Predictability	20.52	331.07	< 0.001
BabyLM 350M	Predictability	22.20	8818.43	< 0.001
BabyLM 1.3B	Predictability	21.83	15067.89	< 0.001

Table 12: GAM tensor-product smooth summary for UP as subword. EDF = estimated degrees of freedom; p = p-value for the te(predictor, layer) smooth term.

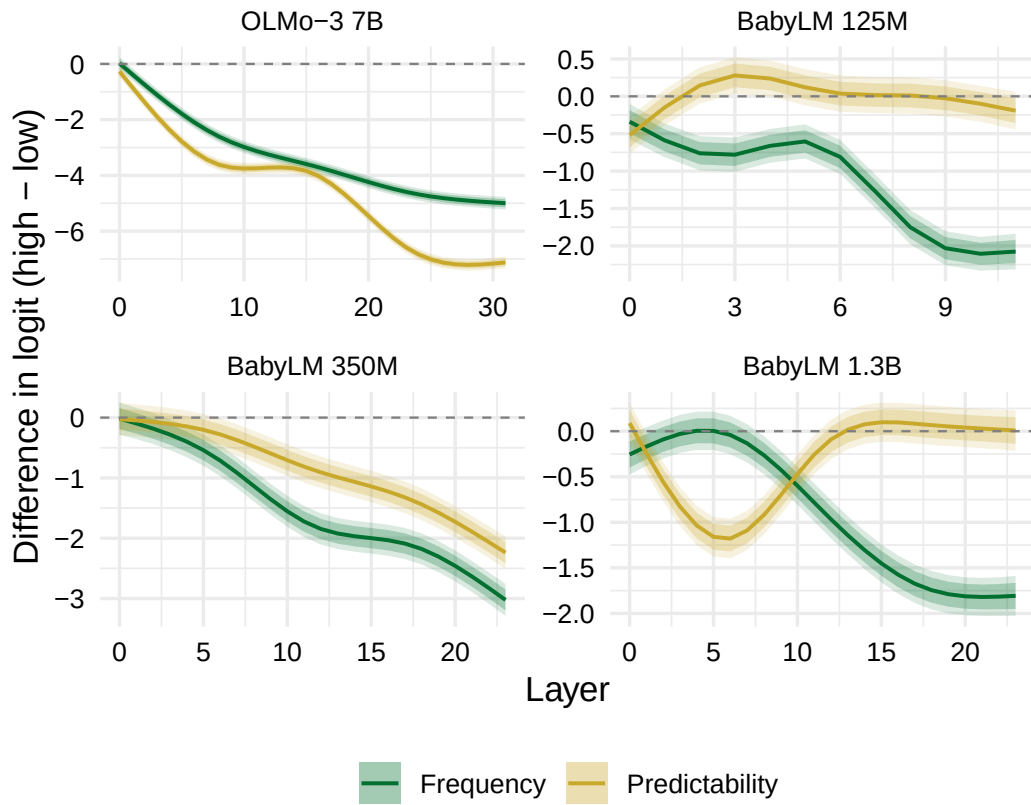


Figure 9: Layer-by-layer difference in GAM-predicted logit between high (90th percentile) and low (10th percentile) predictor values. Green = frequency effect; yellow = predictability effect. Inner and outer shaded ribbons show 80% and 95% confidence intervals derived from the GAM standard error. A negative difference indicates that high-predictor phrases yield a lower classifier logit (i.e., their representation of *up* is less similar to the preposition class).

G Experiment 3: Whisper

The underlying GigaSpeech-derived candidate pool contained 227,898 timestamped speech segments in which *up* occurred, of which 220,285 contained a V+*up* instance, spanning 5,275 unique V+*up* types, before filtering to the final test set described in the main text.

Component	Train pos	Train neg	Val pos	Val neg
Encoder	1,000	1,000	1,000	1,000
Decoder	1,000	1,000	1,000	1,000

Table 13: Classifier training and validation split sizes for Experiment 3 (Whisper). Positive examples are *word_up* instances; negatives are random non-*up* alphabetic tokens from the same segment.

Model	Coef	Est.	Err.	95% CI	% > 0
Encoder	Intercept	9.72	0.04	[9.64, 9.81]	100.0
Encoder	Freq	0.06	0.04	[-0.03, 0.14]	91.0
Encoder	Predic.	-0.07	0.04	[-0.16, 0.02]	5.5
Encoder	Duration	-0.12	0.02	[-0.16, -0.07]	0.0
Encoder	Freq:Predic.	0.01	0.04	[-0.06, 0.09]	64.3
Decoder	Intercept	6.15	0.06	[6.03, 6.28]	100.0
Decoder	Freq	-0.18	0.06	[-0.29, -0.06]	0.1
Decoder	Predic.	-0.70	0.07	[-0.83, -0.58]	0.0
Decoder	Duration	0.07	0.03	[0.01, 0.12]	99.5
Decoder	Freq:Predic.	-0.05	0.06	[-0.16, 0.06]	20.8

Table 14: Joint frequency $\times$ predictability model at the final layer for Whisper encoder and decoder, controlling for the duration of the *up* audio segment (a proxy for phonetic reduction). Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Predictor	EDF	F	p
Encoder	Frequency	13.43	81.33	< 0.001
Decoder	Frequency	21.17	9361.71	< 0.001
Encoder	Predictability	17.39	60.29	< 0.001
Decoder	Predictability	22.29	9258.61	< 0.001

Table 15: GAM tensor-product smooth summary for Whisper encoder and decoder, controlling for the duration of the *up* audio segment. EDF = estimated degrees of freedom; p = p-value for the te(predictor, layer) smooth term.

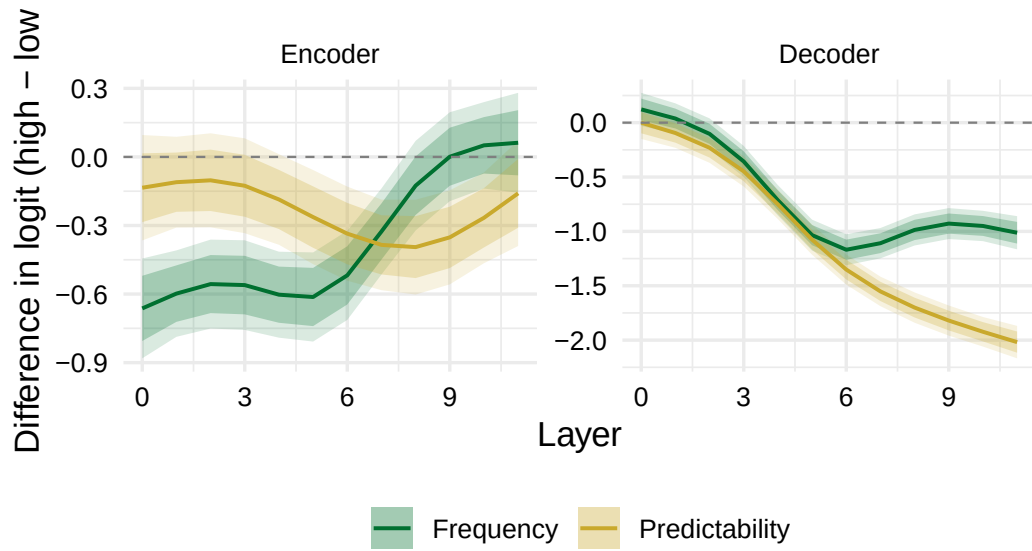


Figure 10: Layer-by-layer difference in GAM-predicted logit between high (90th percentile) and low (10th percentile) predictor values, for Whisper encoder and decoder. Green = frequency effect; yellow = predictability effect. Inner and outer shaded ribbons show 80% and 95% confidence intervals derived from the GAM standard error. A negative difference indicates that high-predictor phrases yield a lower classifier logit (i.e., their representation of *up* is less similar to the preposition class).

H Whisper Subword Replication

As a further robustness check, we replicated Experiment 2’s subword design for Whisper, applying the same up-within-word criteria used there but adapted for audio.

H.1 Methods

H.1.1 Classifier Training

The classifier’s positive training class was broadened to include instances of *up* embedded within a larger word in addition to the standalone (prepositional) *up*. Unlike Experiment 2’s purely orthographic candidate selection, audio candidates were additionally required to actually be *pronounced* with the *up* sound: each candidate word’s CMU Pronouncing Dictionary transcription was checked for a contiguous AH-P phoneme pair (covering both the stressed STRUT vowel, as in *couple*, and the unstressed schwa, as in *support*), which excludes words that merely spell *up* without saying it (e.g., *superintendent*). Since audio has no discrete subword tokens, character-level WhisperX forced alignment was then used to isolate the *up* portion of each host word’s audio span.

Unlike Experiment 2’s text-side design, which draws exactly one instance per unique up-within-word type (a design that works there because roughly 8,662 qualifying types are available in the text corpora), the audio corpus yields far fewer qualifying types, even with a lower floor for inclusion: only a few hundred up-word types meet this bar.

Up-word types were therefore first split into disjoint train and validation sets, so that no type appears in both, and only then were up to five audio instances sampled per type within its assigned split, rather than exactly one. Allowing multiple instances per type is a smaller concession to memorization risk here than it would be for text: because each instance is drawn from a different underlying audio segment and speaker, repeated instances of the same up-word type still give the classifier acoustically distinct embeddings rather than the same feature vector encountered multiple times, unlike, for instance, a token embedding that would recur identically across repeated occurrences of the same word. Negative examples paired with these positives were drawn from the same audio segment and excluded not just the exact word *up* but any word containing *up* as a substring, so a negative could not accidentally be a second up-containing word from the same sentence. Because types themselves are still held disjoint between train and validation, validation accuracy on the subword condition specifically reflects the classifier’s performance on up-within-word types it never encountered during training, rather than repeated exposure to previously-seen words, and is reported separately from validation accuracy on prepositional *up* for exactly this reason (Table 18). The test set is otherwise identical to Experiment 3.

The encoder and decoder draw on the same pool of qualifying up-word types and the same train/validation type assignment, but do not necessarily retain the same instances after embedding extraction. The encoder’s embedding is located directly from audio timestamps, while the decoder’s requires locating the specific token containing *up* in the model’s own tokenization of the transcript; for some subword instances no single decoder token cleanly contains *up* as a substring, and that instance is dropped for the decoder while still being usable for the encoder. Split sizes are therefore reported separately per component rather than assumed to match.

H.1.2 Analyses

Predictor definitions are otherwise identical to Experiment 3. Following Experiment 3’s own design, both the joint model at the final layer and the by-layer GAM analyses additionally include the duration of the *up* segment (*c_duration*) as a covariate, to control for the same phonetic-reduction confound (Equation 5 and Equation 6, respectively). Unlike the main-text analyses, we additionally fit single-predictor “frequency alone” and “predictability alone” models (likewise controlling for duration) here, in order to diagnose the joint model’s frequency coefficient patterns in both components, discussed in Appendix H.3 below.

H.2 Results

Two patterns depart from the main experiment, for different reasons. The decoder’s joint-model frequency coefficient reverses sign relative to the rest of the paper; as we demonstrate in Appendix H.3 below, this is a

statistical-suppression artifact of the frequency-predictability correlation, not evidence that the underlying (negative-leaning) frequency effect fails to replicate. The encoder’s frequency coefficient is also positive here, but for a different reason: it is credible both in the joint model and in the single-predictor model alone (Table 16), so it is not a suppression artifact but a genuine, reproducible positive effect, a stronger version of the same directional, though non-credible, pattern already present in the main experiment’s encoder. Predictability’s effect, by contrast, replicates as negative in both components throughout, exactly as in the main experiment.

Split sizes are given in Table 17. Per-layer validation accuracy, broken down into overall (combined standalone-*up* + subword-*up* positives) and subword-only accuracy, is given in Table 18. Full numerical results are reported in Table 16 (single-predictor models), Table 19 (joint model), and Table 20 (GAM); the individual (single-predictor) model effects are visualized in Figure 11 and Figure 12, with Figure 13 showing the layer-by-layer difference directly.

Considered on their own (Table 16), the two predictors show different patterns by component, controlling for duration as in the joint model. In the decoder, both frequency and predictability lean negative, matching the pattern found throughout the paper: predictability is credible, and frequency is negative-leaning (95.8% of the posterior below zero) though just short of the conventional 95% credibility threshold. It is this negative-leaning frequency effect, not a genuine positive one, that the joint model’s sign flip (discussed below) obscures. In the encoder, predictability is likewise negative-leaning (94.5% below zero) but frequency is different: it is credibly *positive* on its own (100.0% of the posterior above zero), matching the sign and credibility of the joint-model coefficient reported below. Because the alone and joint models agree here, this cannot be a suppression artifact, unlike the decoder’s frequency coefficient, where alone and joint disagree.

Model	Coef	Est.	Err.	95% CI	% > 0
Encoder	Freq. (alone)	0.09	0.03	[0.04, 0.15]	100.0
Encoder	Predic. (alone)	-0.04	0.03	[-0.09, 0.01]	5.5
Decoder	Freq. (alone)	-0.09	0.05	[-0.18, 0.01]	4.2
Decoder	Predic. (alone)	-0.40	0.04	[-0.48, -0.31]	0.0

Table 16: Single-predictor “frequency alone” and “predictability alone” models for the Whisper subword replication, each also controlling for duration (*c_duration*) as in the joint model. Estimates are from Bayesian mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

In the joint model, which follows Experiment 3’s design in also including *c_duration* as a covariate (Equation 5), predictability’s coefficient remained credible in both components and was, if anything, larger than in the main experiment. Frequency’s encoder coefficient replicates the main experiment’s direction. In the decoder, the effect of frequency diverges from that of the main experiment, with its CI crossing zero and its sign flipping to positive. This sign flip is not due to a lack of an effect, but rather due to statistical suppression, which we demonstrate in the following section.

Component	Train pos	Train neg	Val pos	Val neg
Encoder	1,775 (775)	1,775	1,775 (775)	1,775
Decoder	1,686 (686)	1,686	1,696 (696)	1,696

Table 17: Classifier training and validation split sizes for the Whisper subword replication. Positive examples combine 1,000 prepositional *up* instances (as in Experiment 3) with up to five instances per up-within-word type; parenthetical counts give the subword-specific portion. Negatives are matched 1:1 using the same criteria as Experiment 3.

Layer	Encoder		Decoder	
	Overall	Subword	Overall	Subword
0	94.8	91.0	97.7	90.9
1	95.8	93.0	98.0	91.5
2	97.0	94.8	98.2	92.7
3	97.5	96.0	98.4	93.5
4	98.0	96.9	97.9	91.5
5	98.5	97.4	97.8	91.2
6	98.7	97.9	97.9	92.0
7	98.7	98.2	98.0	93.1
8	98.8	97.9	97.7	92.2
9	99.0	97.5	97.1	90.2
10	99.0	97.8	95.7	87.6
11	99.2	98.2	94.5	84.1

Table 18: Held-out validation accuracy (%) by layer and component for the Whisper subword replication. Overall = accuracy on the full combined validation set (standalone-*up* + subword-*up* positives vs. negatives); Subword = accuracy restricted to the subword-*up* validation rows only, i.e. up-within-word types never seen during training (see discussion above). The main text’s aggregate validation-accuracy claim (Appendix C) uses the Overall column; the Subword column is reported here to make the type-generalization check described above concrete.

Model	Coef	Est.	Err.	95% CI	% > 0
Encoder	Intercept	11.98	0.05	[11.88, 12.07]	100.0
Encoder	Freq	0.19	0.05	[0.10, 0.28]	100.0
Encoder	Predic.	-0.14	0.05	[-0.23, -0.04]	0.3
Encoder	Duration	-0.10	0.03	[-0.16, -0.05]	0.0
Encoder	Freq:Predic.	-0.03	0.05	[-0.11, 0.06]	28.2
Decoder	Intercept	9.05	0.09	[8.88, 9.21]	100.0
Decoder	Freq	0.05	0.08	[-0.11, 0.22]	74.1
Decoder	Predic.	-0.78	0.09	[-0.95, -0.60]	0.0
Decoder	Duration	0.08	0.04	[-0.01, 0.17]	96.7
Decoder	Freq:Predic.	-0.12	0.08	[-0.28, 0.04]	6.8

Table 19: Joint frequency $\times$ predictability model at the final layer for the Whisper subword replication, controlling for the duration of the *up* audio segment (Equation 5). Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Predictor	EDF	F	p
Encoder	Frequency	7.38	289.17	< 0.001
Decoder	Frequency	20.42	1144.50	< 0.001
Encoder	Predictability	13.43	132.55	< 0.001
Decoder	Predictability	21.79	1184.45	< 0.001

Table 20: GAM tensor-product smooth summary for the Whisper subword replication, encoder and decoder, controlling for the duration of the *up* audio segment (Equation 6), matching Experiment 3’s own by-layer GAM design. EDF = estimated degrees of freedom; p = p-value for the te(predictor, layer) smooth term.

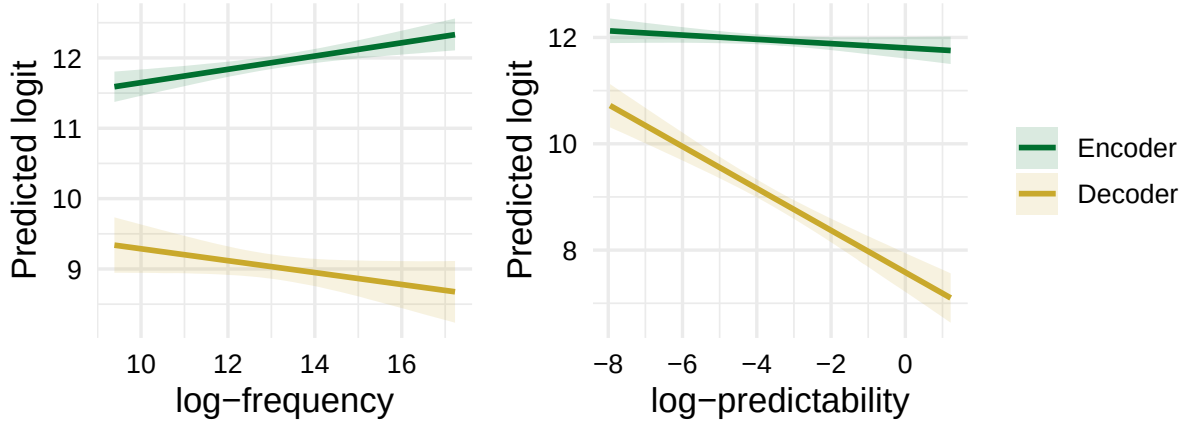


Figure 11: Predicted logit by frequency (left) and predictability (right) for Whisper encoder and decoder, subword replication, from the individual (single-predictor) models rather than the joint model. Shading indicates 95% CI.

H.3 Statistical Suppression

Statistical suppression occurs when two correlated predictors, entered into the same regression, produce a partial coefficient for one of them that differs in sign from its own independent relationship with the outcome, even though neither predictor’s independent relationship with the outcome is itself in question (Friedman and Wall, 2005). This is not a sign of an unreliable model: the standard collinearity diagnostic, the variance inflation factor (VIF), is 1.10 for this predictor pairing, comparable to every other analysis in this paper (1.07–1.13 across all model families). VIF depends only on how correlated the two predictors are with each other, not on how strongly each is independently related to the outcome, so it cannot itself flag which specific analyses are at risk of this pattern.

When two predictors are entered into the same standardized regression, the least-squares partial coefficient of whichever predictor has the smaller zero-order correlation with the outcome is

$$\hat{\beta}_{\text{weak}} = \frac{r_{y,\text{weak}} - r_{y,\text{strong}} \cdot r_{1,2}}{1 - r_{1,2}^2} \quad (7)$$

where $r_{y,\text{weak}}$ and $r_{y,\text{strong}}$ are the two predictors’ zero-order correlations with the outcome and $r_{1,2}$ is the correlation between the predictors themselves (Friedman and Wall, 2005). The denominator is always positive, so the sign of $\hat{\beta}_{\text{weak}}$ depends entirely on the numerator, which turns negative exactly when

$$r_{1,2} > \frac{r_{y,\text{weak}}}{r_{y,\text{strong}}} \quad (8)$$

Table 21 computes both sides of this inequality for every model and condition in the paper. The predictor correlation $r_{1,2}$ is fixed by the shared test set and is identical across the indep and subword conditions for a given model. Three of the ten conditions have an observed $r_{1,2}$ exceeding this threshold: all three Whisper conditions except the decoder under independent training. Everywhere else, the same background correlation stays below each condition’s threshold, so those coefficients can only attenuate, never reverse.

Exceeding the threshold, however, only flags a condition as *at risk* of suppression; it cannot by itself distinguish a real effect being suppressed from a case where the weaker predictor’s own relationship is simply too close to zero for the linear approximation to be informative either way. We adjudicate this directly using the single-predictor (alone) models, and the three flagged Whisper conditions turn out to fall into three different categories. In the subword encoder, the alone-model frequency coefficient is credibly *positive* (Table 16), agreeing with the joint model rather than contradicting it; there is nothing to suppress, and we treat this as a genuine, replicating positive effect. In the indep encoder, by contrast, neither the alone-model estimate (81.4% of the posterior above zero) nor the joint-model estimate (91.0%, Table 14) reaches credibility in either direction; this is not suppression either, since there is

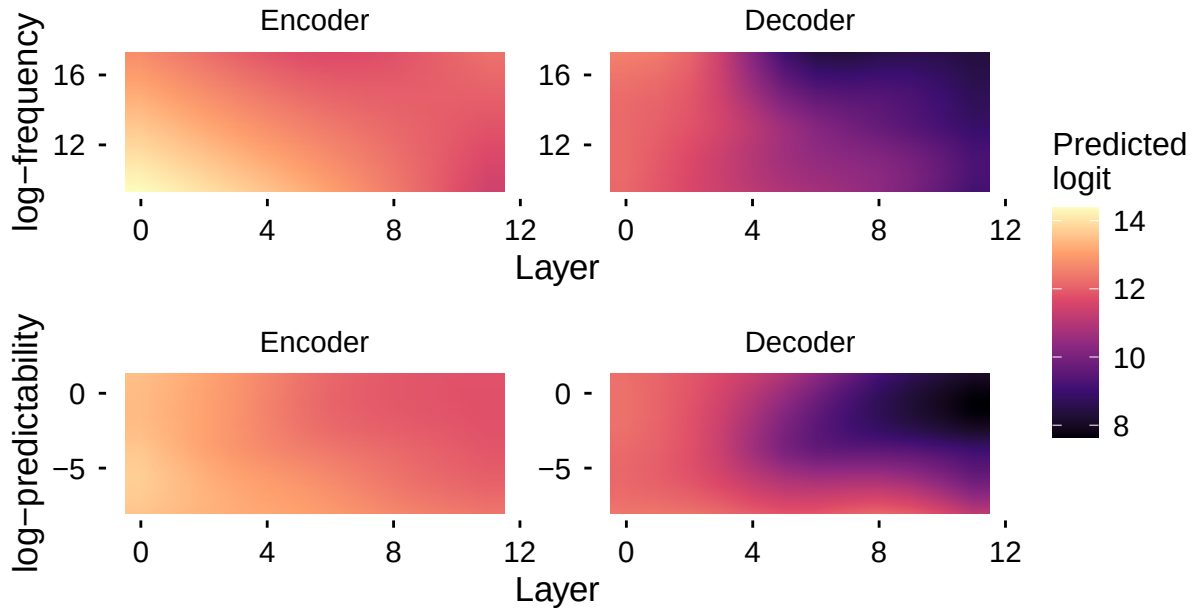


Figure 12: GAM-predicted logit by layer and predictor for Whisper encoder and decoder, subword replication. Top: frequency; bottom: predictability. Color = logit; darker = lower logit (less similar to the preposition *up*); random effects excluded.

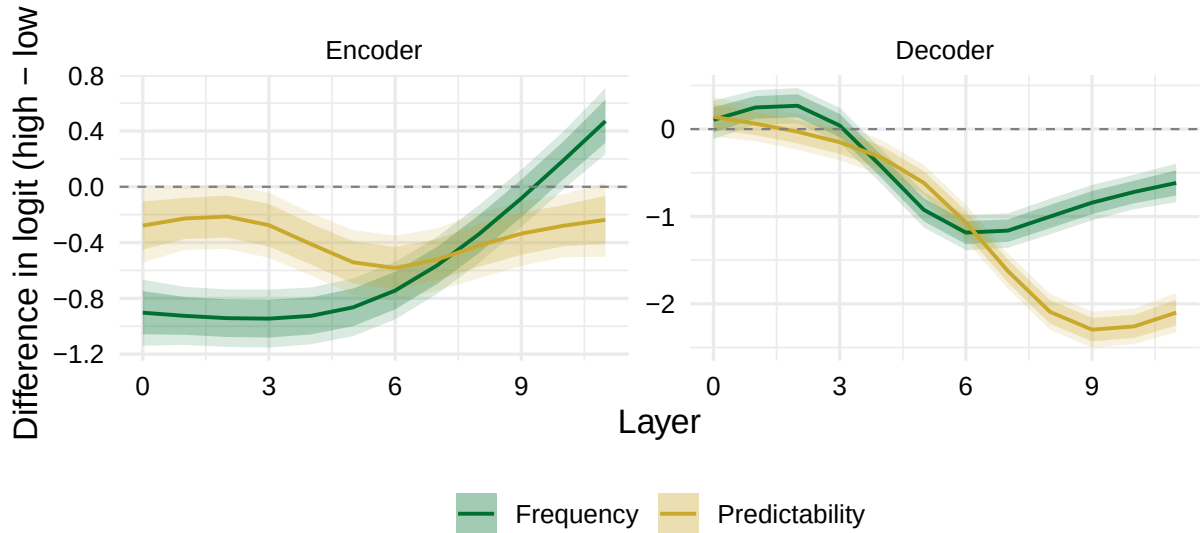


Figure 13: Layer-by-layer difference in GAM-predicted logit between high (90th percentile) and low (10th percentile) predictor values, for Whisper encoder and decoder, subword replication. Green = frequency effect; yellow = predictability effect. Inner and outer shaded ribbons show 80% and 95% confidence intervals derived from the GAM standard error. A negative difference indicates that high-predictor phrases yield a lower classifier logit (i.e., their representation of *up* is less similar to the standalone, preposition class).

no credible effect on either side to be masked or revealed, so we treat frequency in the indep encoder as not meaningfully different from zero. Only the decoder-subword condition shows the alone/joint disagreement that suppression specifically predicts: alone frequency is negative-leaning (95.8% of the posterior below zero, matching frequency’s sign in every other condition in the paper), while the joint model is non-credible and positive-leaning (Table 16, Table 19). We therefore reserve the suppression explanation for the decoder-subword condition alone.

Condition	$r(Y, \text{freq})$	$r(Y, \text{predic})$	$r_{1,2}$	Threshold	Result
OLMo indep	-0.338	-0.539	0.337	0.628	stable
OLMo subword	-0.294	-0.438	0.337	0.672	stable
BabyLM-125M indep	-0.287	-0.159	0.251	0.554	stable
BabyLM-125M subword	-0.212	-0.057	0.251	0.267	stable
BabyLM-350M indep	-0.297	-0.100	0.251	0.338	stable
BabyLM-1.3B indep	-0.316	-0.119	0.251	0.375	stable
Whisper encoder indep	0.013	-0.017	0.306	-0.748	sign flip
Whisper encoder subword	0.039	-0.018	0.306	-0.465	sign flip
Whisper decoder indep	-0.079	-0.193	0.306	0.411	stable
Whisper decoder subword	-0.022	-0.129	0.306	0.168	sign flip

Table 21: Threshold is the right-hand side of Equation 8 for that condition; Result indicates whether the observed $r_{1,2}$ exceeds it. Two BabyLM subword conditions (350M, 1.3B) are omitted: predictability’s zero-order correlation with the logit is already near zero there prior to any joint modelling, a distinct phenomenon not addressed by this mechanism.

Equation 8 is derived for a simple two-predictor regression on standardized variables; our fitted models additionally include a frequency-by-predictability interaction and a random intercept for verb, and are estimated in a Bayesian rather than least-squares framework. Recomputing the decoder’s frequency coefficient with each of these added incrementally shows the sign is unaffected by any of them: the coefficient remains positive in the subword condition and negative in the indep condition at every stage, though its magnitude shifts somewhat under the random-intercept specification. The threshold values in Table 21 should therefore be read as illustrating the mechanism rather than as an exact algebraic derivation of the reported posterior estimates.

An alternative explanation is that frequency’s negative zero-order correlation is itself an artifact of its correlation with predictability, and the joint model reveals frequency’s true, non-negative relationship rather than suppressing a real one. Suppression requires two things at once: the predictors must be correlated, and the weaker predictor’s own relationship to the outcome must be small enough for that correlation to swamp it. The first condition holds equally across all ten conditions in this paper ($r_{1,2} = 0.251$ – 0.337), so it cannot by itself explain why any particular condition reverses. Comparing the magnitude of frequency’s zero-order correlation across conditions is not decisive here either, since all three flagged Whisper conditions have small values (-0.022 to 0.039) and yet only one is genuine suppression, as established directly above via the alone-model comparison. That comparison, rather than the magnitude of the zero-order correlation alone, is what distinguishes the decoder-subword condition (alone and joint disagree) from the two encoder conditions (alone and joint agree, whether both credible or both non-credible).

Taken together: predictability replicates as a negative effect throughout; the decoder’s frequency departure has a specific, well-understood statistical explanation (suppression) rather than reflecting a substantive non-replication; and the encoder’s frequency effect is not a departure from the main experiment at all, but a credible version of the same small positive lean already present there. We therefore consider the subword replication to parallel Experiment 3’s results throughout, including in the decoder.